

Final Project Proposal: Bayesian Estimation of Co-Occurrence Risk

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Scenario

The recent near-collision incident has prompted a fishing pause and highlighted the critical need for accurate estimation of fish co-occurrence probability. The Ministry has conducted a limited aerial surveillance study during the fishing pause: over 20 days, the two species were observed in the same area on 7 days.

The Ministry's historical belief, derived from the transition matrix analysis in Assignment 8, estimates the long-run co-occurrence probability at $P(\text{Same}) = 0.345$. This project will use Metropolis-Hastings MCMC to combine this prior knowledge with the new surveillance data to obtain a complete posterior distribution for the true co-occurrence probability, ψ . This posterior distribution will quantify the current uncertainty and provide a robust basis for deciding whether to lift the fishing pause.

Notation

- ψ : The unknown true probability of co-occurrence on any given day. ($0 \leq \psi \leq 1$)
- k : Number of successful co-occurrence observations ($k = 7$)
- n : Total number of observational trials ($n = 20$)
- $p(\psi)$: The prior distribution representing historical knowledge. We will use a Beta distribution centered on the Assignment 8 result.
- $p(k \mid \psi, n)$: The likelihood function, which is $\text{Binomial}(n, \psi)$.
- $p(\psi \mid k, n)$: The posterior distribution we wish to estimate.

Proposed Model

We adopt a simple yet powerful Bayesian model. The likelihood of observing k co-occurrence events in n trials is given by the Binomial distribution:

$$p(k \mid \psi, n) = \binom{n}{k} \psi^k (1 - \psi)^{n-k}.$$

We encode the historical knowledge from Assignment 8 ($P(\text{Same}) = 0.345$) as a Beta prior distribution, $p(\psi) = \text{Beta}(\alpha, \beta)$. The parameters α and β are chosen so that the mean of the prior is 0.345, reflecting our historical belief.

The resulting posterior distribution is proportional to the product of the likelihood and prior:

$$p(\psi \mid k, n) \propto p(k \mid \psi, n) \cdot p(\psi).$$

This posterior is analytically tractable (it is another Beta distribution), but we will deliberately use the Metropolis-Hastings algorithm to approximate it. This provides a perfect controlled environment to learn, implement, and validate the MCMC method on a simple one-dimensional problem.

Proposed Analysis

The analysis will proceed as follows:

1. Define the Binomial likelihood and Beta prior based on the historical value from Assignment 8.
2. Implement the algorithm to sample from the posterior distribution $p(\psi \mid k, n)$.
3. Assess the reliability of the MCMC samples using trace plots and the Gelman-Rubin statistic (running multiple chains).
4. Summarize the posterior distribution for ψ by calculating its mean, median, and 95% credible interval.
5. Compare the MCMC approximation to the known analytical posterior solution to verify the implementation is correct.

Expected Outcome

This project will deliver:

- A functioning and well-understood Metropolis-Hastings sampler for a fundamental Bayesian inference problem.
- A posterior distribution for ψ that combines historical and new data, providing a complete picture of risk and uncertainty.

- A direct comparison between computational (MCMC) and analytical solutions
- A clear recommendation to the Ministry on whether the posterior probability of co-occurrence remains unacceptably high.